

Applied Statistics and Inference Assignment

Question 1: What are Type I and Type II errors in hypothesis testing, and how do they impact decision-making?

Answer:

Type I Error occurs when we reject a null hypothesis that is actually true. It is also called a false positive. For example, a medical test may indicate that a healthy person has a disease.

Type II Error occurs when we fail to reject a null hypothesis that is actually false. It is also called a false negative. For example, a medical test may indicate that a sick person is healthy.

Both errors affect decision-making. Type I errors can lead to unnecessary actions and costs, while Type II errors can cause missed opportunities or serious consequences. Researchers try to balance these risks by choosing an appropriate significance level.

Question 2: What is the P-value in hypothesis testing, and how should it be interpreted in the context of the null hypothesis?

Answer:

The P-value measures how likely the observed results are if the null hypothesis is true. A small P-value indicates strong evidence against the null hypothesis.

If the P-value is less than the chosen significance level (usually 0.05), the null hypothesis is rejected. Otherwise, there is not enough evidence to reject it.

The P-value does not prove whether the null hypothesis is true or false; it only helps us evaluate the strength of evidence against it.

Question 3: Difference between a Z-test and a T-test.

Answer:

A Z-test is used when the population standard deviation is known and the sample size is large. A T-test is used when the population standard deviation is unknown, especially for smaller samples.

The Z-test uses the normal distribution, while the T-test uses the t-distribution, which accounts for additional uncertainty in smaller samples.

Question 4: What is a confidence interval, and how does the margin of error influence its width and interpretation?

Answer:

A confidence interval is a range of values that is likely to contain the true population parameter. It is usually written as: Sample Estimate $\pm$ Margin of Error.

A larger margin of error creates a wider confidence interval, while a smaller margin of error creates a narrower interval. Wider intervals provide more certainty but less precision, whereas narrower intervals provide more precision but less certainty.

Question 5: Describe the purpose and assumptions of an ANOVA test.

Answer:

ANOVA (Analysis of Variance) is used to compare the means of three or more groups at the same time. Its main purpose is to determine whether at least one group mean differs significantly from the others.

Assumptions: 1. Observations are independent. 2. Data follows a normal distribution. 3. Variances are approximately equal across groups.

ANOVA extends hypothesis testing beyond two groups, avoiding the need for multiple t-tests and reducing the risk of Type I errors.

Question 6: Write a Python program to perform a one-sample Z-test.

Answer:

```
import numpy as np
from scipy.stats import norm

data=[52,55,50,53,54,56,58,57,55,54]
mu=50
sigma=5

x_bar=np.mean(data)
n=len(data)
z=(x_bar-mu)/(sigma/np.sqrt(n))
p=2*(1-norm.cdf(abs(z)))
print(z,p)
```

Interpretation: If the p-value is below 0.05, reject the null hypothesis.

Question 7: Simulate a dataset from a binomial distribution (n=10, p=0.5) using NumPy and plot the histogram.

Answer:

```
import numpy as np
import matplotlib.pyplot as plt

data=np.random.binomial(10,0.5,1000)
plt.hist(data,bins=11)
plt.show()
```

This code generates 1000 observations from a binomial distribution and displays their frequency distribution.

Question 8: Generate multiple samples from a non-normal distribution and implement the Central Limit Theorem using Python.

Answer:

```
import numpy as np
import matplotlib.pyplot as plt

means=[]
for i in range(1000):
    sample=np.random.exponential(scale=2,size=30)
    means.append(np.mean(sample))

plt.hist(means,bins=30)
plt.show()
```

Although the original distribution is not normal, the distribution of sample means becomes approximately normal, demonstrating the Central Limit Theorem.

Question 9: Write a Python function to calculate and visualize the confidence interval for a sample mean.

Answer:

```
import numpy as np
import scipy.stats as stats

def confidence_interval(data):
    mean=np.mean(data)
    ci=stats.t.interval(0.95,len(data)-1,
        loc=mean,scale=stats.sem(data))
    print(ci)
```

The function calculates a 95% confidence interval and can be extended with a graph for visualization.

Question 10: Perform a Chi-square goodness-of-fit test using Python.

Answer:

```
from scipy.stats import chisquare
```

```
observed=[20, 30, 25, 25]
```

```
expected=[25, 25, 25, 25]
```

```
chi2, p=chisquare(observed, expected)
```

```
print(chi2, p)
```

If the p-value is less than 0.05, the observed distribution differs significantly from the expected distribution. Otherwise, there is no significant difference.